Example 5.6

Is there any mistake in the data provided about the two regression lines $Y = -1.5\,X + 7$, and $X = 0.6\,Y + 9$? Give reasons.

Solution:

The regression coefficient of Y on X is $b_{YX} = -1.5$

The regression coefficient of X on Y is $b_{XY} = 0.6$

Both the regression coefficients are of different sign, which is a contrary. So the given equations cannot be regression lines.

Example: 5.7

	mean	S.D
Yield of wheat (kg. unit area)	10	8
Annual Rainfall (inches)	8	2

Correlation coefficient: 0.5

Estimate the yield when rainfall is 9 inches

Solution:

Let us denote the dependent variable yield by Y and the independent variable rainfall by X.

Regression equation of Y on X is given by

$$Y - \bar{y} = r_{XY} \frac{SD(Y)}{SD(X)} (x - \bar{x})$$

$\bar{x} = 8,\ SD(X) = 2, \bar{y} = 10,\ SD(Y) = 8,\quad r_{XY} = 0.5$

$$Y - 10 = 0.5 \times \frac{8}{2} (x - 8)$$

$$= 2\,(x - 8)$$

When $x = 9$,

$$Y - 10 = 2\,(9 - 8)$$

$$Y = 2 + 10$$

$$= 12 \text{ kg (per unit area)}$$

Corresponding to the annual rain fall 9 inches the expected yield is 12 kg (per unit area).

Example 5.8

For 50 students of a class the regression equation of marks in Statistics (X) on marks in Accountancy (Y) is $3Y - 5X + 180 = 0$. The mean marks in of Accountancy is 50 and variance of marks in statistics is $\frac{16}{25}$ of the variance of marks in Accountancy.

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